

Original Research

Functional Efficacy of Full Balanced Occlusion and Lingualized Occlusion Evaluated from Morphological Differences in the Edentulous Alveolar Ridge

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Clinical significance

Although both full balanced occlusion and lingualized occlusion are occlusion types applied to complete dentures, there are no definitive clinical guidelines for their selection. In this study, suggested guidelines were evaluated from morphological differences in the residual alveolar ridge.

ABSTRACT

Purpose : This study examined the functional efficacy of complete dentures with full balanced occlusion or lingualized occlusion in patients with different degrees of residual ridge resorption.

Materials and Methods : To compare the residual ridge condition among the five subjects, the alveolar ridge height and supporting area under the denture were measured. Experimental upper and lower complete dentures were fabricated with interchangeable molars and were used by the study subject for two months prior to the functional analyses. To estimate denture function, electromyography (EMG), mandibular kinesiology (MKG), and pressure distribution under the denture while tapping and masticating peanuts were investigated.

Results : The EMG results suggested that subjects with poor residual ridge condition may benefit from lingualized occlusion for more efficient mastication. The MKG results suggested that, in subjects with poor ridge condition, mastication tends to be unstable with a full balanced denture, but that widening of the mandibular path that destabilizes the denture is reduced with a lingualized denture. The results of pressure measurement suggested that when lingualized occlusion is provided for patients with a poor residual ridge condition, the occlusal pressure is transmitted mostly at the lingual side of the residual ridge, leading to efficient denture function.

Conclusion : A sufficient lingualized occlusion effect was obtained in subjects with a poor alveolar ridge condition. The following indexes were examined as the criteria for applying lingualized occlusion. The ratio of the residual alveolar ridge must be 0.5 or less, and the area of the supporting base of the denture must be 2,000 mm² or less.

Key words

lingualized occlusion, full balanced occlusion, functional evaluation, electromyography, pressure distribution

This article was presented at the first Meeting of the Asian Academy of Prosthodontics, April 1999, Seoul, Korea.

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Received on April 12, 2002/Accepted on July 1, 2002

INTRODUCTION

Functional recovery through the use of complete dentures markedly depends on the condition of the alveolar ridge supporting the masticatory pressure. In patients with a notably resorbed alveolar ridge, sufficient retention and support of dentures are not obtained, causing masticatory disorders and pain. The occlusal schemes of complete dentures for those patients include full balanced occlusion, lingualized occlusion, and monoplane occlusion. Lingualized occlusion is based on the concept that the biting force applied to dentures is always guided in the direction of the lingual side of the alveolar ridge to stabilize the denture base, which physiologically fits the jaw movement. Since in the occlusion the upper cusp come into contacts at points in the lower fossa, the biting force necessary for crushing and masticating food is relatively small, which protects the alveolar ridge.

There have been many reports on lingualized occlusion since Pound¹⁾ proposed the theory of this type of occlusion in 1970. Its functional usefulness in comparison with full balanced occlusion and monoplane occlusion has recently been reported²⁻¹⁴⁾, and occlusal schemes were mechanically analyzed with regard to their relationship with occlusal surface morphology and food characteristics^{3,6,9,10-14)}. However, there have been few reports on the functional usefulness of this occlusion in clinical cases^{3,11,12)}. Functional usefulness compared to full balanced occlusion has been clinically evaluated in some reports in which masticatory values and electromyograms were measured in clinically applied cases of lingualized occlusion. It was found that this occlusion was useful for patients with alveolar ridge resorption; however, the relationship between the degree of alveolar ridge resorption and the morphology of the alveolar ridge has not

been investigated.

In this study, the denture function of full balanced occlusion and of lingualized occlusion was compared in the same subject, and among five subjects with different degrees of alveolar ridge resorption in order to evaluate the morphological conditions of the alveolar ridge, and thus to maximize the usefulness of lingualized occlusion.

MATERIALS AND METHODS

Study Subjects

The study subjects participating in the study were recruited from patients who were treated with upper and lower complete dentures at the Tsurumi University School of Dental Medicine Affiliated Hospital. Five subjects exhibiting different degrees of residual ridge resorption were included in this study (Fig. 1). To compare the residual ridge condition among the subjects, the lower alveolar ridge height and the lower denture base area were measured. The alveolar ridge height was measured following the method by Wical et al.¹⁵⁾ and Shiina et al.¹⁶⁾ The ratio of the alveolar ridge height at the mental foramen on panoramic radiographs was measured. The distance from the lower border of the mental foramen to the upper border of the mandible was expressed as, a, the distance from the lower margin of the mental foramen to the lower border of the mandible as, b, and the height of the residual alveolar ridge as a/b. The supporting area under the mandibular complete denture was also measured by a three-dimensional surface coordinator (Grasp, Techno Arts, Tokyo, Japan).

Experimental Complete Denture

Experimental upper and lower complete dentures were fabricated, including interchangeable molars with screw bolts, which allowed the

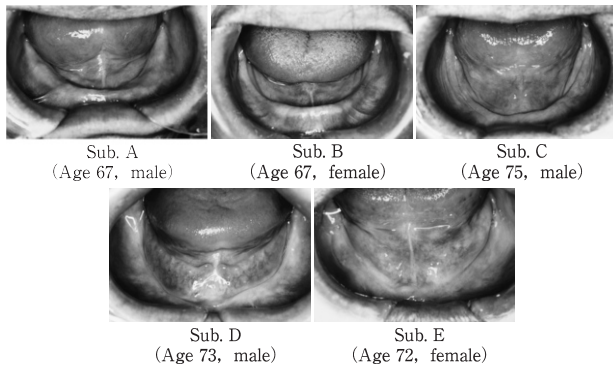


Fig. 1 The edentulous mandibular alveolar ridge in five subjects

comparison of 20-degree resin molars (Bio-ace, Shofu, Kyoto, Japan) for full balanced occlusion (FBO) and lingualized molars (GC, Tokyo, Japan) for lingualized occlusion (LO) (Fig. 2). Both dentures were bilaterally balanced.

Functional Analyses

Evaluation of function was performed after the subjects had used the experimental FBO dentures for two months. The same evaluation was performed after the subjects had used the experimental LO dentures for the subsequent two months.

The functional evaluation was performed by first testing using electromyography (EMG) of the masticatory muscles, then mandibular kinesiography (MKG), and finally examination of pressure distribution under the lower denture. Electromyograms of the medio-inferior area of the bilateral masseter muscles and the anterior area of the bilateral temporal muscles were taken using dipolar surface electrodes (E/RE, NEC-Sanei, Tokyo, Japan). The distance between the bilateral electrodes was 20 mm in the masseter muscle, and 15 mm in the temporal muscle. The muscle activity potential induced from the surface electrodes was amplified using a bio-electrical phenomenon amplifier (180-4 system, Biophysigraph type 1205 D, NEC-Sanei, Tokyo, Japan). Using these electromyograms,

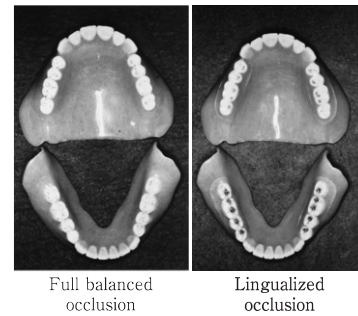


Fig. 2 Experimental upper and lower complete dentures

Only the artificial molar teeth were removable. Twenty-degree resin molars (Bio-ace, Shofu, Kyoto, Japan) were used for full balanced occlusion, and specific resin molars (Lingualized, GC, Tokyo, Japan) were used for lingualized occlusion. Bilateral balanced occlusion was applied to both occlusion types

the chewing stroke, total integrated muscle activity, and integrated muscle activity per chewing stroke were measured during the mastication and swallowing of 2.0 g of peanuts.

The mandibular movement was monitored by MKG (K 6-I Diagnostic System, Myotronics, Seattle, WA, USA) while tapping and masticating 2.0 g of peanuts; each width of the sagittal and frontal mandibular movement paths was measured at the 5.0 mm opening position from the intercuspal position. To attach the K 6-I magnet to the lower denture, the labial area of the artificial anterior teeth of the lower denture was partly ground, so that the denture would not move due to protrusion of the magnet.

The distribution of pressure locally applied to the alveolar ridge under the denture base was measured while clenching the teeth and while masticating peanuts. After producing duplicates of the FBO and LO experimental dentures and adjusting them in the oral cavity, pressure sensors were attached to the buccal and lingual sides of the basal surface of the denture, and between the areas under the second premolar and under the first molar, using the position of the



Fig. 3 Basal surface view of the duplicated lower denture, to which the pressure sensor was attached (left) ; the measurement of applied pressure (right)

The sensor attached the right buccal side was sensor #1, to the right lingual side sensor #2, to the left lingual side sensor #3, and to the left buccal side sensor #4

lower artificial teeth as the reference. A pressure sensor (PS-10 KA, Kyowa Electric, Tokyo, Japan), with a maximum tolerable pressure of 10 kg/cm², was directly attached to the surface of the duplicate denture base using acrylic resin (Fig. 3). The gap between the flat pressure-applied surface of the sensor and mucosal surface of the alveolar ridge was filled with silicone (Fit-checker, GC) so that equal pressure could be applied to the pressure point of the sensor. Before measuring the pressure, the experimental denture was placed in the oral cavity for five minutes for temperature correction of the sensor, in which a zero point compensation was performed without applying pressure to the denture. The occlusal pressure generated after five strokes from the beginning of mastication was recorded for 15 seconds, and the integrated pressure was calculated.

EMG, MKG, and pressure applied locally to the alveolar ridge were measured five times in each subject, and the mean value of the five measurements was calculated. Statistical comparisons between the occlusal schemes were performed using a paired *t*-test at a 5% level of significance.

RESULTS

The morphological conditions of the residual alveolar ridge were compared among the subjects. Figure 4 shows the relationship between the height of the lower alveolar ridge and the area of the lower denture base in each patient. The ratio of the alveolar ridge height ranged from 0.25 to 1.22, and the area of the supporting base of the dentures ranged from 1,574 mm² to 3,687 mm². Among the five subjects, the alveolar ridge condition was best in subject A, showing the maximum ratio of alveolar ridge height to maximal area. The values decreased in the order of B, C, D, E. The alveolar ridge condition was poorest in subject E, showing approximately 1/5 of the ratio of the alveolar ridge height, and approximately 1/2 of the area in subject A. The subjects were classified based on the ratio of the alveolar ridge height, according to the report by Shiina et al.¹⁶⁾, in which alveolar ridge height was measured using the same method as this study. The ridge condition was classified as relatively satisfactory in Subjects A, B, and C, and as relatively poor in Subjects D and E. Clinically, the height and width of the alveolar ridge were sufficient, and the ridge condition was judged to be good in Subjects A and B. In Subjects D and E, however, the ridge condition was extremely poor, and they were clinically regarded as difficult-to-treat cases.

Figure 5 shows the results of the EMG measurements. The minimum number of masticatory strokes between the beginning of mastication and swallowing was 17.3, and the maximum number was 47.7 in FBO. In LO, the minimum number was 23.2, and the maximum number was 36.9. The frequency of mastication was further increased in subjects with poorer alveolar ridge condition. The frequency of mastication was smaller in FBO than in LO in

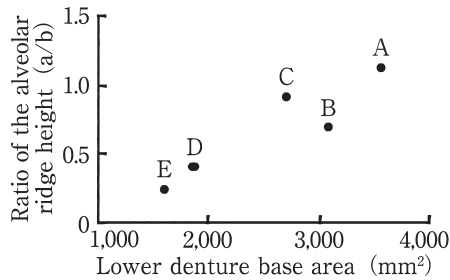


Fig. 4 Residual ridge condition of the subjects Relationship between the ratio of the mandibular alveolar ridge height and the lower denture base area in each subject

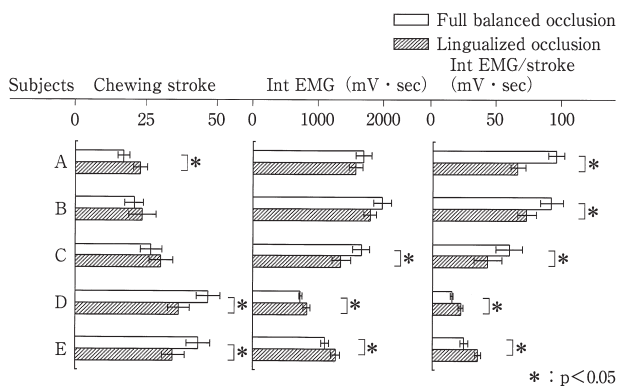


Fig. 5 EMG analysis during the period from mastication to swallowing 2.0 g of peanuts

patients A, B, and C with good alveolar ridge condition, whereas the frequency was significantly larger in FBO than in LO in patients D and E with a poor alveolar ridge condition ($p < 0.05$).

The total amount of muscle activity was large in subjects with good alveolar ridge condition, compared to subjects with poor alveolar ridge condition. The total amount of muscle activity decreased more in FBO compared to LO in patients with good alveolar ridge condition, whereas it was greater in FBO than in LO in patients with poor alveolar ridge condition.

The total amount of muscle activity per chewing stroke was significantly larger ($p < 0.05$) in FBO than in LO in subjects with good alveolar ridge condition, whereas it was significantly larger ($p < 0.05$) in LO than in FBO in subjects with poor alveolar ridge condition.

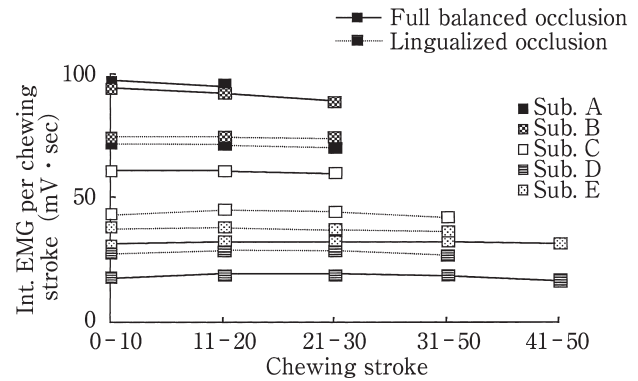


Fig. 6 Integrated EMG per chewing stroke for each 10 stroke period

The mean value of the integrated EMG per 10 strokes between the start of mastication and swallowing 2.0 g of peanuts

Each mean value of the total amount of muscle activity per 10 strokes is shown in Fig. 6. The total amount of muscle activity per stroke was larger in subjects with better alveolar ridge condition, whereas it was smaller in those with a poorer condition. There were no significant differences among each mean value per 10 strokes between the start of mastication and swallowing in both FBO and LO.

The results of the MKG measurements are shown in Fig. 7. During tapping, the width of the mandibular movement path in the sagittal and frontal segments was further increased in subjects with poorer alveolar ridge condition. The width of the LO movement path was significantly narrower ($p < 0.05$) in the frontal segment in subject D, and in the sagittal and frontal segments in subjects A and E in comparison with FBO. No significant differences were noted between FBO and LO in subjects B and C. Also, while masticating peanuts, the width of the movement path was further increased in subjects with poorer alveolar ridge condition. Although the width of the movement path in subjects D and E was less in LO than in FBO, significant differences were present.

The results of the measurement of locally

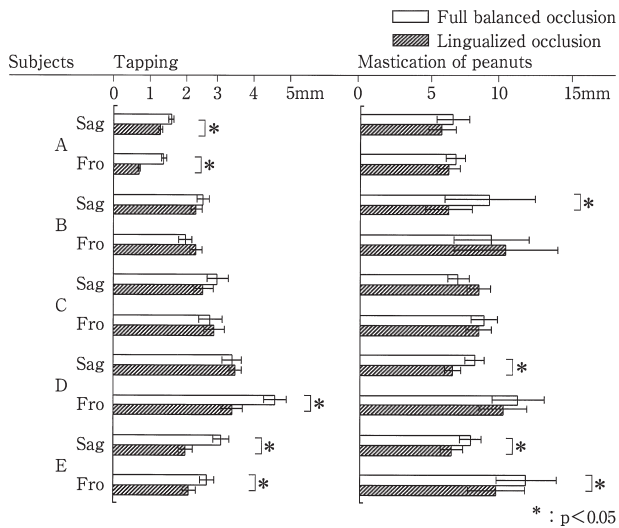


Fig. 7 The width of the mandibular movement path during tapping and mastication of peanuts

The mandibular path width at the sagittal and frontal segments when the mouth was opened 5 mm from the intercuspal position were measured

applied pressure showed increasing rates of applied pressure in LO in comparison with FBO (Fig. 8). In subjects A and B with good alveolar ridge condition, no differences in applied pressure were found between FBO and LO during both clenching and mastication. Although no differences in applied pressure were observed between FBO and LO during clenching in subjects D and E with poor alveolar ridge condition, the applied pressure was significantly higher ($p < 0.05$) on the lingual side than on the buccal side during mastication in LO.

DISCUSSION

In complete denture cases with highly advanced alveolar ridge resorption, it is very difficult to achieve retention and support of dentures due to decreases in residual alveolar ridge height and width. Lingualized occlusion can be selected as an approach for treating these difficult cases. The reasons for selecting lin-

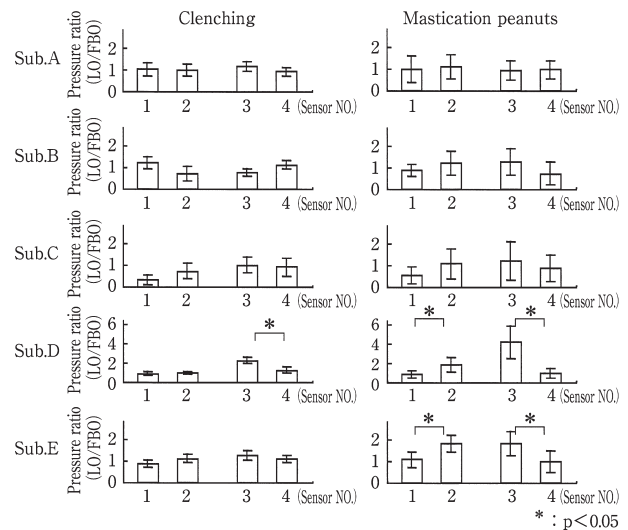


Fig. 8 The ratio of pressure applied to the alveolar mucosa under the LO denture base to that under the FBO denture base

Pressure ratio was indicated as the ratio of pressure applied to the mucosa under the LO denture to that under the FBO denture in each position of the pressure sensor

gualized occlusion are : 1) the biting force applied to dentures is always guided in the direction of the lingual side of the alveolar ridge, acting in a direction to stabilize the denture base, 2) the occlusal contact relationship is simple and is harmonized with masticatory function in comparison with that in full balanced occlusion, and 3) since food crushability is readily adjusted by changing the occlusal surface morphology, the burden on the residual alveolar ridge can be decreased, which therefore protects it.

The functional usefulness of lingualized occlusion was reported in the 1980 s and 1990 s. Evaluations of the distribution of pressure applied to the bucco-lingual sides of the residual alveolar ridge indicated that the rate of applied pressure apparently increased on the lingual side in lingualized occlusion, showing that the biting force acted in a direction to stabilize the lower dentures. Lingualized occlusion is, therefore, more appropriate for the support and retention of upper dentures than full balanced occlusion^{5,6,10}.

After mechanically analyzing the relationship between occlusal schemes and occlusal surface morphology or food characteristics, it has been reported that the force necessary for mastication can be controlled by changing the occlusal surface morphology, and efficient mastication was obtained by small masticatory force in lingualized occlusion^{3,6,9,10-14}. By evaluating the masticatory performance and electromyograms in clinically applied cases of lingualized occlusion, researchers found that lingualized occlusion showed higher functionality than full balanced occlusion from the perspective of increased masticatory performance and protection of the alveolar ridge by stabilizing the dentures^{3,11,12}. Although the application of full balanced occlusion to complete denture cases was not necessarily ruled out in these reports, it is certain that there have been no clear criteria for choosing between full balanced occlusion and lingualized occlusion. Koide et al.¹⁷ reported the criteria applicable to clinical cases, which showed the application methods for various cases regarding tooth arrangement and modification. Matsumoto et al.¹⁸ suggested guidelines for selecting posterior artificial teeth for complete dentures. Although lingualized occlusion can be applied to partial and complete dentures in a wide range of applications, the application criteria to maximize the usefulness of lingualized occlusion from the perspective of the morphological condition of the edentulous alveolar ridge have not yet been reported. Therefore, it is clinically important to compare denture function¹⁹ by changing only the occlusal scheme of the same denture in the same subject, as in this study, and then to determine the application criteria based on the alveolar ridge condition under which the effect was observed.

Five patients with different alveolar ridge resorption conditions were selected as subjects. Generally, the difficulty in complete denture cases is greatly related to the morphological

condition of the residual alveolar ridge. Therefore, as indexes of the morphological condition of the residual alveolar ridge, the alveolar ridge height and denture-supporting area were selected for investigation^{20,21}. Measuring the ratio of edentulous mandibular bone height using the same method as this study, Shiina et al.¹⁶ reported that the mean ratio in 122 subjects was 0.66 on the right side and 0.69 on the left side. Since subjects A, B, and C in this study showed higher values than the mean ratio, it was determined that they had a comparatively high alveolar ridge. On the other hand, subjects D and E showed lower values than the mean ratio, so it was determined that they had a comparatively low alveolar ridge. Measuring the basal surface area of lower denture bases using the same method as this study, Kawasaki²² reported that the mean value in 49 subjects was 2,317 mm², with a maximum value of 4,566 mm² and a minimum value of 1,211 mm². Since subjects A, B, and C in this study showed higher values than the mean value, it was determined that they had a comparatively wide denture-supporting area, whereas subjects D and E had a comparatively narrow denture-supporting area because they showed values lower than the mean value. From these two results, it was determined that the alveolar ridge condition was good in subjects A, B, and C, whereas it was poor in subjects D and E.

With regard to the experimental dentures, only the artificial teeth were changed. To observe changes in denture function due to differences in occlusal schemes, it is necessary to change only the occlusal scheme in the same denture base in the same subject, by which accurate measurement is possible without the influence of other factors. Regarding the period necessary to become accustomed to the dentures and occlusal scheme, the subjects used the dentures for two months after occlusal adjustment.

During this period, no adjustment of the basal surface of the denture base was performed. In general, studies to evaluate the differences in occlusal schemes based on electromyograms and masticatory efficiency are performed using patients, whereas studies to evaluate the differences in occlusal schemes based on supporting pressure of the alveolar ridge under the denture base are performed using simulation models. It is important to note that both of the present studies were performed using patients under the same condition.

The EMG results showed a general tendency toward a further increase in the frequency of mastication in subjects with poorer alveolar ridge condition. The frequency of mastication in FBO decreased in comparison with LO in subjects A, B, and C with a good alveolar ridge condition, and it significantly decreased in LO in subjects D and E with a poor alveolar ridge condition. These results suggested that efficient mastication is possible in subjects with a poor alveolar ridge condition by applying LO, whereas masticatory efficiency decreased in subjects with a good alveolar ridge condition by applying LO. Further support for these results was that the total amount of muscle activity per stroke was high for FBO in subjects with a good alveolar ridge condition and was also high for LO in subjects with poor alveolar ridge condition. Measuring masticatory performance using the same frequency of mastication as in this study, Gomibuti et al.¹¹⁾ reported that there were no differences in masticatory performance between FBO and LO while eating peanuts. Since our study was based on the electromyographical approach, easy comparison of the results was impossible; however, it was considered that no differences in masticatory performance occurred in subjects C and D with an intermediate alveolar ridge condition. The calculation of each mean total amount of muscle activity per 10 strokes

showed that there were no differences in muscular working efficiency between FBO and LO during the period between the start of mastication and swallowing. This finding suggested that even though the occlusal schemes differed, the chewing method shifted in the same way in both FBO and LO during the period between the start of mastication and swallowing.

The MKG results showed that the width of the movement path was further increased in subjects with poorer alveolar ridge condition, showing a larger variation of the movement path. This suggested the direct influence of jaw movement stability including the mobility of denture base. It was considered that the movement path was the most stable when occlusal schemes appropriate for the alveolar ridge condition in each subject were established. The width of the movement path in LO was significantly narrow ($p < 0.05$) in the frontal segment in subject D and in the sagittal and frontal segments in subjects A and E, whereas no significant differences were noted between FBO and LO in subjects B and C. Studies¹¹⁾ to evaluate changes in the masticatory cycle during masticating food between FBO and LO, using the same methodology as this study, showed that the stability of the masticatory cycle was higher in LO than in FBO. Although there were different viewpoints regarding the masticatory cycle and masticatory path, both methods showed the same result: changes in the masticatory path in LO decreased in subjects with a poor alveolar ridge condition. On the other hand, although no significant differences were found between FBO and LO in subjects with a good alveolar ridge condition, a tendency toward increases in the width and changes was observed in FBO. These findings suggested that mastication in a stable movement path with slight changes is possible by applying LO for subjects with a poor alveolar ridge condition, whereas the effect of applying LO was unclear in

subjects with a good alveolar ridge condition.

The evaluation of functional differences between FBO and LO regarding the distribution of pressure applied to the alveolar ridge under denture bases has mainly been performed using simulation models of alveolar mucosa^{5-10,13,14}; there have been few reports of examinations performed on human alveolar mucosa because the leads of the pressure sensors attached to experimental dentures markedly inhibited mastication. Measurements in human alveolar mucosa were made possible in this study by simplifying the structure of the experimental dentures and by simplifying the data; pressure applied on only the buccal and lingual sides was measured, and the amount of pressure applied for 15 seconds halfway through mastication was compared. No subject showed any difference in the bucco-lingual ratio of supporting pressure between FBO and LO during clenching. Performing maxillary simulation model experiments, Matsumoto et al.⁶ reported that differences in occlusal schemes did not influence the distribution of supporting pressure in centric occlusion, as was also found in this study. Supporting pressure significantly increased on the lingual side in comparison with the buccal side during mastication in subjects with a poor alveolar ridge condition in LO, which agreed with the results of many previous reports. On the other hand, no clear effect of LO was obtained in subjects with a good alveolar ridge condition.

It was considered that, since sufficient supporting area of the alveolar ridge and alveolar ridge height was present in subjects with a good alveolar ridge condition, differences in the distribution of bucco-lingual supporting pressure due to differences in the morphology of the occlusal surface did not readily occur in the subjects. It was thought that the anterior movement of the lower denture during occlusion would be large in patients with severe alveolar ridge

resorption. Therefore, it was speculated that the pressure applied to the sensor on the lingual side would increase when this propellant phenomenon occurred. Although it was impossible to identify the presence of the propellant phenomenon from the research data, we determined that lingualized occlusion occurred, based on several factors: 1) increases in lingual pressure were noted in experiments by Inoue et al.⁵, Matsumoto et al.⁶ and Ohguri et al.⁹; 2) the width of the antero-posterior path was smaller in LO than in FBO as measured by MKG; and 3) neither pressure spots nor decubitus formed on the lingual side, which was a good sign clinically, while the subjects used the LO dentures for two months. These findings suggested that the effect of lingualized occlusion differed according to the alveolar ridge condition.

CONCLUSION

Functional comparisons between full balanced occlusion and lingualized occlusion were performed on subjects with different alveolar ridge conditions. As a result, no differences in functionality due to differences in occlusal schemes were noted in subjects with a good alveolar ridge condition. A sufficient LO effect was obtained in subjects with a poor alveolar ridge condition, suggesting that efficient mastication could be attained. The following indexes were examined as the criteria for applying lingualized occlusion. First, the ratio of the residual alveolar bone should be less than approximately 0.5, in which the distance between the inferior border of the mental foramen and the superior border of the mandibular bone should be less than half the distance between the inferior border of the mental foramen and the inferior border of the mandibular bone. Second, the supporting area of the denture base should be less than approximately 2,000 mm², showing the

cord-shaped alveolar ridge covered with movable mucosa. It was suggested that patients could perform sufficient mastication with a small biting force by applying lingualized occlusion under these criteria.

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